

Multitask CapsNet: An Imbalanced Data Deep Learning Method for Predicting Toxicants

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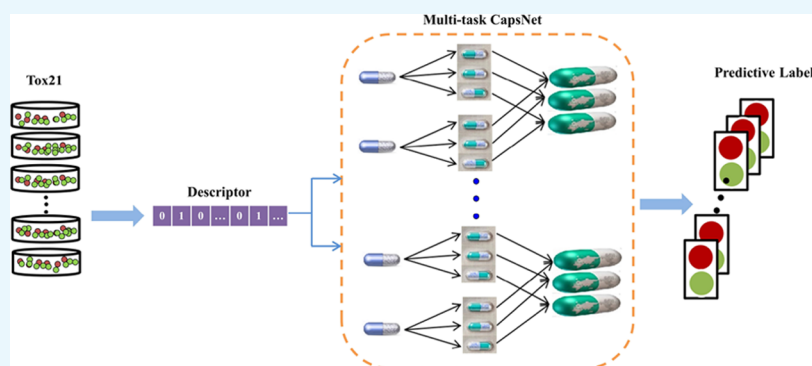
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ABSTRACT: Drug development has a high failure rate, with safety properties constituting a considerable challenge. To reduce risk, *in silico* tools, including various machine learning methods, have been applied for toxicity prediction. However, these approaches often confront a serious problem: the training data sets are usually biased (imbalanced positive and negative samples), which would result in model training difficulty and unsatisfactory prediction accuracy. Multitask networks obtained significantly better predictive accuracies than single-task methods, and capsule neural networks showed excellent performance in sparse data sets in previous studies. In this study, we developed a new multitask framework based on a capsule neural network (multitask CapsNet) to measure 12 different toxic effects simultaneously. We found that multitask CapsNet excelled in toxicity prediction and outperformed many other computational approaches using the multitask strategy. Only after training on biased data sets did multitask CapsNet achieve significantly improved prediction accuracy on the Tox21 Data Challenge, which gave the largest ratio of highest accuracy (8/12) among compared models. Our model gave a prediction accuracy of 96.6% for the target NR.PPAR γ , whose ratio of negative to positive samples was up to 36:1. These results suggested that multitask CapsNet could overcome the bias problems and would provide a novel, accurate, and efficient approach for predicting the toxicities of compounds.

INTRODUCTION

Although good progress has been made in drug research and development (R&D) technology, about 90.4% of compounds entering clinical trials end in failure, with 24.0% failing due to a lack of safety.^{1,2} Distinguishing toxicants from nontoxicants is a decades-old problem with important implications for drug discovery, especially for triaging the toxicities of compounds before proceeding with further optimization. Usually, numerous tests *in vitro* and *in vivo* are performed to explore drug safety, especially different toxicities. Although various toxicity tests are implemented throughout the whole process of R&D, toxicity may vary with the individuals, leading to failure or even market withdrawal.^{3–5} Furthermore, using experimental methods to test toxicity usually requires the use of animal models; and animal testing is not only time-consuming and costly but also involves ethical issues. For these reasons, *in silico* approaches for predicting the toxicities of compounds have become a research hotspot.^{6–9} Compared with experimental methods, the computational approaches have

advantages including speed, cost, accuracy, and environmental benefits.^{10–13} More importantly, computational approaches can be performed even before the compounds are synthesized. Therefore, various computational models have been developed to predict toxicities and have shown great efficiency.^{14–18}

In recent years, with the continuous improvement of computing power, rapid development of machine learning theory, and accumulation of toxicity data libraries, computational methods for predicting toxicity have achieved unprecedented development. The development of deep learning (DL) has been especially significant, with more and

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more high-precision toxicity prediction models established by various novel DL algorithms. Lai's group has developed state-of-the-art DL models to predict different toxicities, such as acute oral toxicity and drug-induced liver injury.^{14,19} Furthermore, researchers have successfully applied multiple DL algorithms to predict drug-induced cardiotoxicity.^{20–22} These models are established for a certain type of toxicity, but multiple toxicities are required to test in drug R&D. Therefore, it is necessary to develop multitask prediction models to predict multiple toxicities simultaneously and improve efficiency greatly. Moreover, previous studies reported that a mass of multitask models obtained higher predictive accuracy than single-task methods under the same conditions.^{23–27} As one of the widely used data sets for developing toxicity prediction models, Tox21 contains more than 12 000 different compounds, and it possesses 12 different toxic effects.^{28,29} Thus, Tox21 is a reasonable choice as a benchmark data set for building multitask toxicity prediction models. Previously, Mayr et al. employed a deep neural network (DNN) to develop a multitask model (DeepTox) based on the Tox21 data set and demonstrated that DeepTox possessed excellent performance against many other traditional methods, including naive Bayes (NB), support vector machines (SVMs), and random forests (RF).³⁰ Subsequently, many multitask models based on DL algorithms have been applied to predict various toxicities. Utilizing a graph-based convolutional neural network (CNN), Coley et al. developed a multitask model to predict toxicities and found that the performance of their model was comparable to DeepTox.³¹ Zaslavskiy et al. combined three algorithms with various molecular descriptors to establish a novel multitask model for toxicity prediction that outperformed existing state-of-the-art models.³² Abbasi et al. proposed a novel multitask architecture; this approach achieved excellent performance by transferring related knowledge from the source to the target data set.³³

Although these multitask DL models based on Tox21 have surpassed traditional machine learning methods in most targets, many challenges remain.^{34,35} For example, the multitask DNN did not perform as good as the traditional machine methods for 5 out of 12 toxicities, suggesting that continuous improvement is required for multitask DL models.³⁰ This problem was related to the commonly used DL algorithms, which require massive data to train models. Moreover, Tox21 is severely imbalanced; the volume of the inactive (negative/nontoxic) data set is much larger than that of the active (positive/toxic) data set. As a result, multitask deep learning models are unable to thoroughly explore the essence of the minority class data set consisting of the positive compounds.^{36,37} To address this issue, several data augmentation approaches have been introduced in previous studies, such as the resampling effect and the synthetic minority over-sampling technique (SMOTE).^{38,39} Manual preprocessing is an essential ingredient of these data augmentation strategies, which may impact the objectivity and performance of the models to some extent. Data augmentation technologies are also not suitable for direct use in chemicals. In addition, sampling weight and class weight are also conventional measures to cope with imbalanced data. However, these measurements may change the range of the loss and affect the stability of the training.⁴⁰ The above approaches required tedious data preparation, which may be time-consuming and costly.

Recently, we proposed novel drug molecular design methods based on CapsNet and applied them to drug cardiotoxicity prediction, drug-induced carcinogenicity prediction, active compound screening, and other toxicity predictions. These novel methods performed well in small-sized sparse data sets.^{41,42} Furthermore, we found that dynamic routing ensures that the information of the low-level capsule is maximally preserved in the high-level capsule. So, compared with other DL architectures, CapsNet can retrieve finer information of molecular features and could efficiently learn more characteristics even if a certain class of sample is limited. This provides a unique opportunity to address the imbalanced data problem without manually engineered features. In this investigation, we continue to apply CapsNet to develop a new multitask learning model, multitask CapsNet, through the Tox21 Data Challenge.^{28,29} Multitask CapsNet outperformed five other machine learning models for predicting all 12 toxicities.³⁰ Due to the unique dynamic routing algorithm of CapsNet, as well as the advantages of the multitask models, multitask CapsNet could automatically mine the effective information for the minority class in imbalanced data sets, thereby improving the overall prediction accuracy. The obvious novelty aspect of this study is that we established a multitask classification model for predicting 12 different toxic effects using modified CapsNet architecture, and by training on imbalanced data, the new model performed better than others without manually engineered data and features.

■ RESULTS AND DISCUSSION

Optimization of Multitask CapsNet. To find the best multitask capsule network architecture for the prediction of the 12 different toxic effects, a series of multitask models constructed with different numbers of restricted Boltzmann machine (RBM) operations and capsule layers were tested. To ensure a fair comparison, all of the optimizations of the different architectures were accomplished under identical experimental conditions. The performance of models was monitored by the fivefold cross-validation method on the training sets. To avoid deviation caused by random variables, cross-validation of each architecture was repeated 10 times. We used five indicators to assess performance. The prediction results are listed in Table S1. The multitask capsule network architecture consisting of two RBM operations, one Primary-Caps layer, and one DigitCaps layer, achieved the best performance in all five indicators, indicating that this architecture possessed promising potential for further investigation. Although the average sensitivity (SE) values (ranged from 48.2 to 52.8%) of all of the candidate architectures were not quite satisfactory compared with other conventional deep learning models, for these new architectures, the predictive ability of toxicants has been improved greatly. For the imbalanced data set, both the receiver operating characteristic curve (ROC) and Matthew's correlation coefficient (MCC) statistical parameters were regarded as appropriate measures for the quality of binary classifications. As shown in Table S1, the average ROC and MCC values of this architecture were 0.920 and 0.560, respectively. This result further confirmed that the selected architecture was the most suitable model for the Tox21 data set among the five multitask CapsNet architectures.

Fivefold Cross-Validation of Multitask CapsNet. Fivefold cross-validation was used to estimate the predictive ability of the five multitask models (Table S2). To ensure

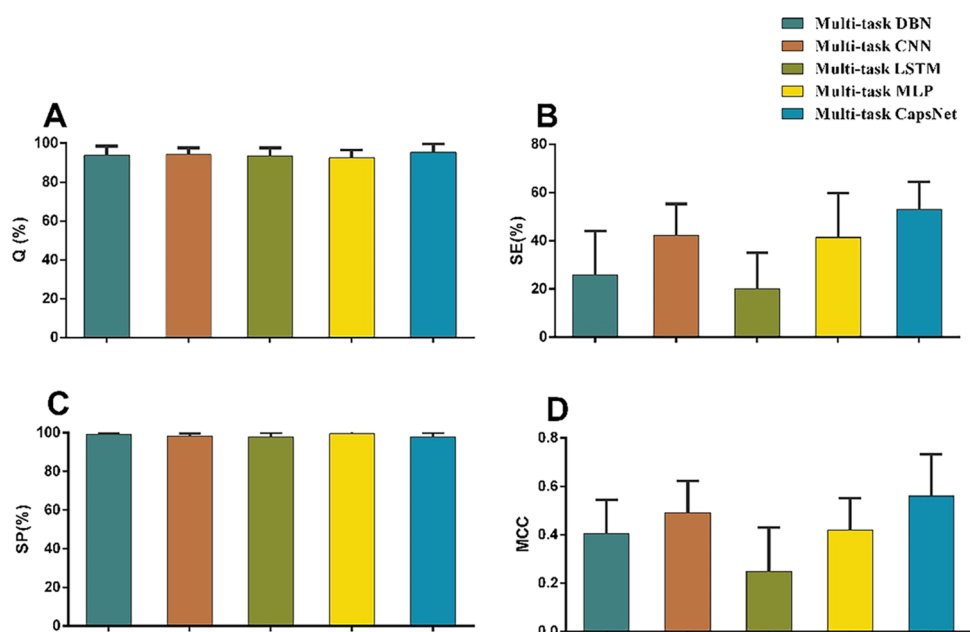


Figure 1. Prediction performance of five multitask models (DBN, CNN, long short-term memory (LSTM), MLP, and CapsNet) on the training data set by fivefold cross-validation. (A) Q, (B) SE, (C) SP, and (D) MCC.

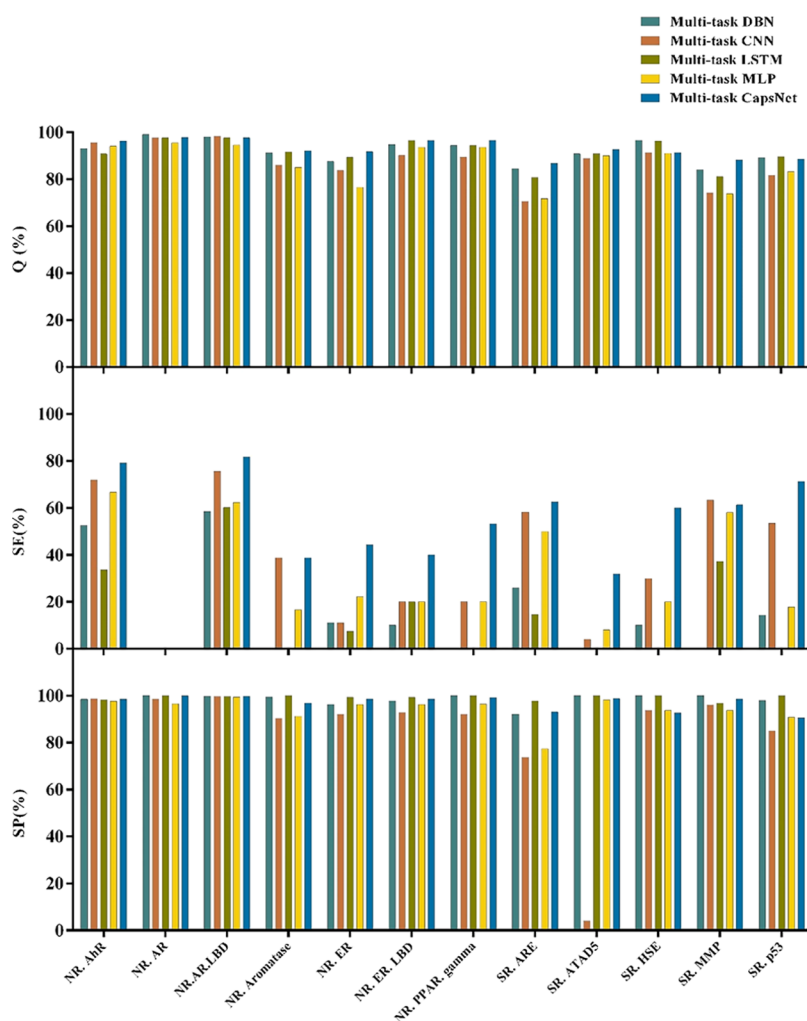


Figure 2. Prediction performance of all multitask models for the 12 separate targets on the leaderboard data set.

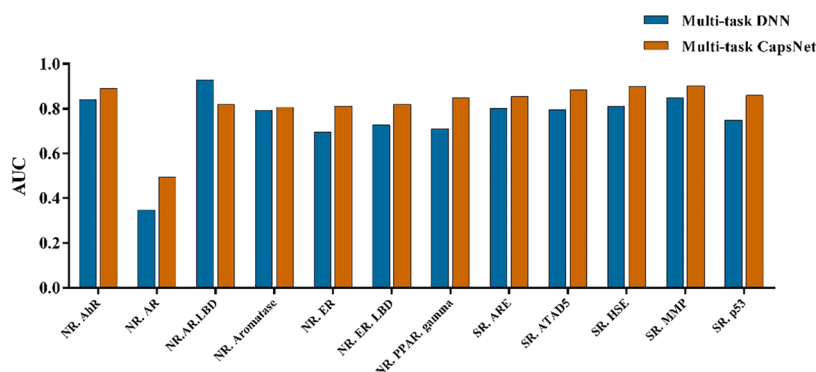


Figure 3. AUC results of multitask CapsNet and multitask DNN in the literature on the leaderboard data set.

unbiased measurement, the random splitting of the training set was repeated 10 times for each model. We used four indicators to assess performance. Among four indicators, multitask CapsNet had three highest values (Figure 1). The overall average predictive accuracies of these five models ranged from 92.3 to 95.3%, with the highest average accuracy of multitask CapsNet of 95.3%. Nevertheless, multitask CapsNet provided the highest sensitivity, ranging from 41.2 to 64.4%. We specifically noted that although with the same architecture for the input layer of multitask CapsNet and multitask deep belief network (DBN), the average sensitivity of multitask DBN was as low as 25.8%, whereas that of multitask CapsNet was as high as 52.8%. For the other models, the average sensitivity values were also relatively lower than that of multitask CapsNet. Although the features of positives are difficult to extract and learn when the number of positive compounds is very limited in a training set, the model's ability to recognize positive compounds greatly improved after the integration of CapsNet and demonstrated that the feature information of fewer compounds can be maintained and amplified well via the routing mechanism in CapsNet. Although the multitask multilayer perceptron (MLP) model had a relatively optimal average specificity (SP) of 99.4%, its Q value was lower than other models. For the average MCC values, the multitask CapsNet model gives the best MCC value of 0.560 to the fivefold cross-validated training set among the five multitask architectures, illustrating that the model developed in this study has excellent robustness and predictive capability for the compounds with new scaffolds. In addition, we compared the multitask CapsNet model with the baseline method on the training data set by fivefold cross-validation. In Table S2, we noticed that our novel model showed more than 2% improvement in terms of both the average accuracy and the SE value and more than 0.05 in terms of the MCC over the best performing baseline, suggesting that the multitask measurement enhanced the predictive capability of the model. The above results showed that the advantage of the multitask CapsNet model in toxic prediction is independent of the number of positive compounds in the training data set.

External Validation of Multitask CapsNet. Tox21 Data Challenge is a severely biased data set, which includes a larger number of negatives compared to positives. Comparing the accuracy of the five models for the 12 targets on external validation, multitask CapsNet gave the largest ratio of highest accuracies (8/12, Figure 2). All of the accuracies of the multitask CapsNet models were greater than 86.8% for the 12 separate prediction targets of the leaderboard data set. The details of the prediction results are listed in Table S3. Multitask

DBN achieved the three highest accuracies, 99.0% (NR.AR), 96.6% (SR.HSE), and 89.2% (SR.p53), but it consumed more than 50 times the number of epochs to reach stability compared with multitask CapsNet. Multitask CNN provided the highest accuracy, 98.4% for the target NR.AR.LBD, with a sensitivity value that was 6.0% lower than that of multitask CapsNet. For the target SR.p53, multitask DBN possessed the highest accuracy (88.5%); however, the sensitivity of multitask CapsNet was higher by 57.1% than that of multitask DBN. For all of the targets, the lowest accuracy was 70.5%, which was provided by the multitask CNN model for the target SR.ARE. Sensitivity for all 12 targets estimated by these models was low due to the high ratio of negatives in the training data sets. The leaderboard data set of each target included a larger number of negatives than positives, and the ratio of positives to negatives ranged from 1:7 to 1:96. In cases like this, predictive models have poor ability for recognizing the positive compounds; thus, sensitivity is usually low. Nevertheless, multitask CapsNet provided the highest sensitivity for 10/12 targets, indicating that this novel model has better ability than others for discriminating positives from negatives even if there are many negatives. For the leaderboard data set of the target NR.AR, which included 4 positives and 249 negatives, multitask DBN's accuracy and specificity were up to 98.0 and 99.7%, respectively, and both were close to those of multitask CapsNet (Q: 98.3% and SP: 99.70%), but the sensitivity of multitask DBN (58.5%) was 23.2% lower than the corresponding value of multitask CapsNet (81.7%).

Multitask CapsNet exhibited good ability to identify a class of compounds of smaller number even when using a biased set to test the model, indicating that multitask CapsNet can learn features from an extremely biased training set. The results are attributed to the excellent generalization ability of the novel model. The dynamic routing algorithm that is the unique advantage of multitask CapsNet ensures that the origin information from the low-level capsule is maximally transmitted and preserved in the high-level capsule. Thus, the multitask CapsNet model could efficiently learn the characteristics of negatives even if their number is limited.

Comparison with Other Models in the Literature. Although many multitask models have been developed for predicting absorption, distribution, metabolism, excretion, and toxicity (ADMET) properties, DeepTox is one of the most recognized multitask models for the Tox21 Data Challenge.³⁰ DeepTox was developed with deep neural networks, which consisted of many hidden layers with thousands of neurons. Mayr et al. indicated that DeepTox outperformed many other computational approaches including naive Bayes, support

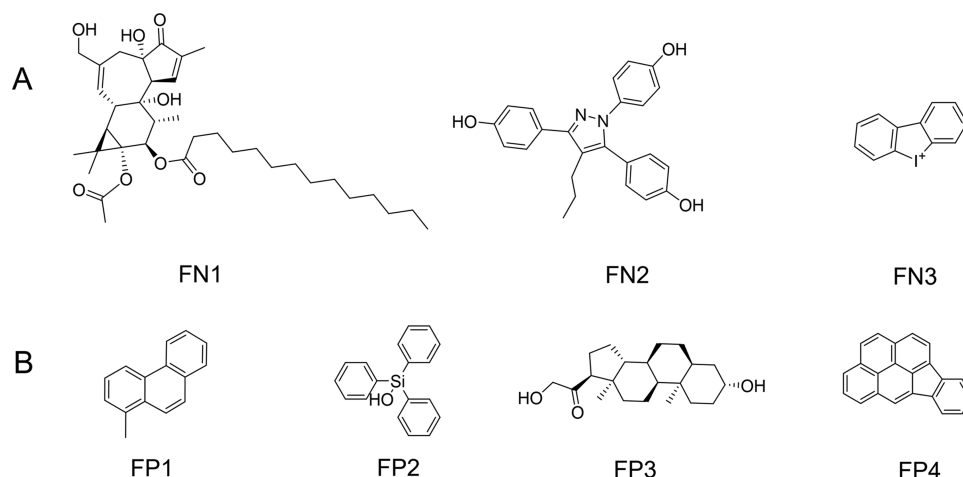


Figure 4. Misclassified compounds of the multitask CapsNet model for the NR.AR.LBD target on the test set, including (A) three false negatives and (B) four false positives.

vector machines, and random forests. Here, we directly compared multitask CapsNet with multitask DNN of the DeepTox pipeline. To make a rigorous comparison, the same data sets were applied to train and test our models. Figure 3 displays the predicted performance of the two models for the leaderboard data set (see detailed information in Table S4). With the same validation set as in Mayr's work, multitask CapsNet showed a better performance than multitask DNN. The results showed that for 11 out of 12 tasks, multitask CapsNet outperformed the multitask DNN model. For the NR.AR.LBD target, the multitask CapsNet model provided a relatively lower area under the curve (AUC, 0.820) in the leaderboard data set. For the rest, the performance of our model was higher by about 10–20% on the corresponding task compared to the multitask DNN of DeepTox. To further estimate the predictive abilities of the multitask CapsNet model, we provided the AUC results over the test set of Tox21, and the performance of our models was compared with the corresponding models from the DeepTox pipeline (Table S5). The multitask CapsNet model provided the best AUC values for 10 out of 12 targets, and DNN and SVM methods provided the highest AUC values for the NR.AR.LBD and SR.MMP targets, respectively. However, in most cases, our models were superior to the compared models in the prediction of two different external validation data sets. These results confirmed the superiority of CapsNet over the traditional DL algorithms, as well as computational approaches for the toxicity prediction model in the literature under the same experimental conditions.

Analysis of Poor Performance for the NR.AR.LBD Target. As discussed above, our novel model developed by CapsNet from the training set yielded the best predictions for most of the targets in two independent test sets, but it still performed slightly poorly for some targets, especially to the NR.AR.LBD target. The possible reasons were that (1) the subnetworks of individual targets obtained fewer sharing features compared with other subnetworks, which may lead to the prediction results being inclined to other tasks with more sharing features, (2) the shared hyperparameter settings of the shared hidden feature layer were not the optimized values for all of the subnetworks, and (3) the limited descriptors were difficult to characterize all of the compounds accurately. Further investigation was carried out for the

structures of seven misclassified compounds (three false negatives and four false positives) in the test set (Figure 4). We found that some scaffolds and structural alerts (SAs) had approximate frequencies in both toxicants and nontoxicants. For example, FN3 and FP1 shared a similar scaffold. In addition, the structures of stereoisomeric molecules (such as FN1 and FP3) were not described in our model, which would significantly influence various properties of compounds. To address this problem, it is necessary to develop more precise three-dimensional (3D) descriptors that can describe the steric positions of atoms and their adjacency relations. At last, due to the low similarity between the false positives of the test set and the positives of the training set, our models could not dig some useful knowledge of positives. By seriously analyzing the Tanimoto similarity indexes for the above compounds, we found that 190 out of 303 positives in the training set provided zero value, and the average value was as low as 0.133.

Analysis of Important Fragments to Toxicants. To better understand important chemical moieties related to the 12 toxic effects, SAs responsible for the 12 types of toxicants were identified by the frequency analysis of substructures from the Klekota–Roth fingerprint (KRFP).⁴³ If a substructure presented more frequently in toxicants than nontoxicants, this substructure would be considered an SA for toxicants. The detailed definition of the “frequency of a fragment” enrichment factor for chemicals is listed in the Supporting Information. In this research, we only analyzed the fragments that appeared more than 10 times in the Tox21 data set.⁴⁴ As shown in Table 1, these fragments identified by the KRFP presented far more frequently in toxicants than in nontoxicants. Therefore, 13 fragments could be regarded as SAs for the 12 types of toxicants. Although important fragment analysis can identify SAs for toxicants, they cannot describe the spatial arrangement of these fragments, or it is difficult to recognize the toxicants if multiple types of fragments are found simultaneously in the same compounds.^{44–46} Once these SAs appear during drug design, careful analysis should be performed.

Comparison of Training Convergence Rates between Multitask CapsNet and Multitask DBN. We further employed the training set of Tox21 to explore the training convergence rate of multitask CapsNet. When using multitask CapsNet and multitask DBN to train this data set, the batch size for both at each training step was 148. The routing

Table 1. Important Fragments for the 12 Types of Toxicants^a

No.	Fragment	Num_P	Freq_P	Num_N	Freq_N	Representative structure
1		34	2.04	9	0.34	
2		40	2.34	4	0.15	
3		30	2.21	5	0.23	
4		26	2.48	1	0.06	
5		20	2.34	2	0.15	
6		20	2.34	2	0.15	
7		15	2.30	4	0.34	
8		18	2.44	1	0.09	
9		17	2.43	1	0.09	
10		14	2.40	1	0.11	
11		11	2.36	1	0.14	
12		11	2.18	2	0.25	
13		10	2.57	0	0.00	

^aAbbreviations: Num_P: the number of positive chemicals, Freq_P: the frequency of a fragment in positive chemicals, Num_N: the number of negative chemicals, and Freq_N: the frequency of a fragment in negative chemicals.

iteration was set to 2 in multitask CapsNet. Figure 5 shows the training loss versus epochs of multitask CapsNet and multitask DBN. As shown in Figure 5, after 1000 epochs of training with multitask CapsNet, the loss converged toward zero, indicating that the model reached stability. By contrast, in training with multitask DBN, the loss just approached 0.45 after more than 18 000 epochs, and this loss value did not noticeably further decline. Apparently, the convergence rate of multitask CapsNet was much faster than that of multitask DBN. Multitask

CapsNet's performance speed was at the level of seconds, while that of multitask DBN was at the level of dozens of minutes. This effect may be attributed to the dynamic routing mechanism in multitask CapsNet. The combinatorial relation between neurons was enhanced by the use of a novel algorithm; thus, the number of parameters could be reduced dramatically in pretraining. With the help of multitask CapsNet, important features can be rapidly focused on, and those with little or no contribution to results can be ignored, and a predictive model with significantly improved performance and fewer epochs can be achieved.

Performance of Various Multitask Models on Balanced Data sets. To further confirm the predictive advantages of multitask CapsNet for the unbalanced data set, we used a series of balanced data sets to train our models. The balanced data sets were selected randomly from the original training data sets of Tox21, and the positive/negative ratio was 1:1 in the new training data sets (Table S6). Table S6 summarizes the prediction results of these two multitask models training on balanced data sets. Multitask CapsNet possesses the same architecture as the multitask DBN model, with the addition of the PrimaryCaps layer and the DigitCaps layer. So, we paid particular attention to the performance difference between these two models. As shown in Table S7, multitask CapsNet showed superior overall performance compared with multitask DBN. The prediction accuracies of multitask CapsNet are in the range of 68.0–89.3%, while those of multitask DBN are in the range of 60.3–76.7%. Comparing the prediction performance of multitask CapsNet training on the balanced data sets with the performance on the unbalanced data sets, the latter was clearly better. When the numbers of capsule neurons from the two types are equal, the crowding of CapsNet may result in the models that are difficult to identify the important features.⁴⁷ As such, multitask CapsNet has unique advantages for unbalanced data sets.

The biased data set is an obstacle to the development of high-precision classifiers, and conventional machine learning methods are of insufficient quality to discriminate positives and negatives simultaneously. As shown in Table 2, the training sets of Tox21 are typical biased data sets, in which the ratio of positives to negatives is as high as 1:36. For fivefold cross-validation, positive elements could hardly be identified using multitask LSTM for both SR.HSE and SR.p53 targets. Multitask CNN yielded good accuracy values for all of the 12 targets, but its MCC scores for some targets were rather low, such as NR.Aromatase. Multitask CapsNet achieved the highest sensitivity and MCC value for each target. In summary, multitask CapsNet, established herein, presented the most accurate and stable performance compared with the most used multitask learning methods.

In our previous studies, the novel drug design approach based on CapsNet showed excellent predictive power and provided a good solution for “small samples” in drug R&D.^{41,42} The “instantiation parameters” of the capsule, which can accurately describe the logical relationships between vector components of the feature space, enable the model to better learn the corresponding molecular features in a small data set. Furthermore, the original CapsNet mainly orients to a two-dimensional image; recent works demonstrated that CapsNet architecture performed well in task prediction using one-dimensional vectors as input, such as natural language processing and bioinformatics.^{48,49} Due to “instantiation parameters”, one-dimensional vectors that possess ambiguous

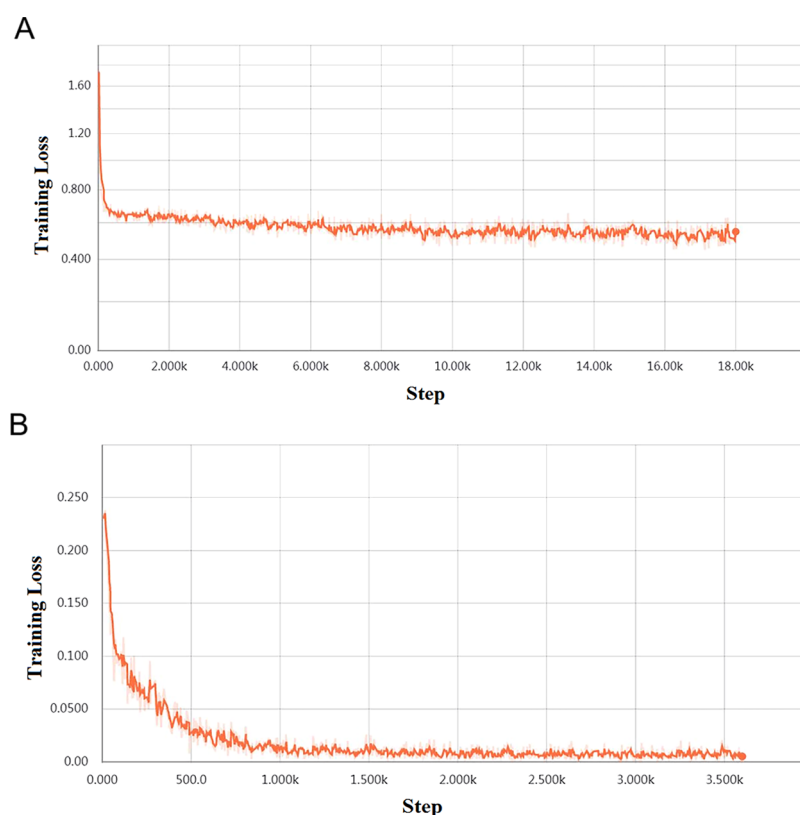


Figure 5. During the fine-tuning stage, the training loss versus epochs of (A) multitask DBN and (B) multitask CapsNet models on Tox21. The horizontal coordinate indicates the training loss values of the two models. The longitudinal coordinate represents the epoch of the two models.

Table 2. Distribution of Positive and Negative Compounds in the Training, Leaderboard, and Test Sets from the Tox21 Data Set

data source		training set			leaderboard set			test set		
		P	N	ratio (P/N)	P	N	ratio (P/N)	P	N	ratio (P/N)
Tox21 Data Challenge	NR.AhR	950	7219	1:8	31	241	1:8	73	537	1:7
	NR.AR ^a	380	8982	1:24	3	289	1:96	12	574	1:48
	NR.AR.LBD ^a	303	8296	1:27	4	249	1:62	8	574	1:72
	NR.Aromatase	360	6866	1:19	18	196	1:11	39	489	1:13
	NR.ER	937	6760	1:7	27	238	1:9	51	465	1:9
	NR.ER.LBD	446	8307	1:19	10	277	1:28	20	580	1:29
	NR.PPAR.gamma ^a	222	7962	1:36	15	252	1:17	31	574	1:19
	SR.ARE	1098	6069	1:6	48	186	1:4	93	462	1:5
	SR.ATAD5 ^a	338	8753	1:26	25	247	1:10	38	584	1:15
	SR.HSE	428	7722	1:18	10	257	1:26	22	588	1:27
	SR.MMP	1142	6178	1:5	38	200	1:5	60	483	1:8
	SR.p53	537	8097	1:15	28	241	1:9	41	575	1:14

^aFour severely biased training sets, and their ratios of positive to negative compounds is larger than 1:20.

logical relationships can be described more accurately. For the current drug design, molecules and proteins are mainly characterized by one-dimensional information, which are suitable for research using the CapsNet algorithm. Not only for toxicity prediction, CapsNet also has great potential in other drug-related prediction tasks. One of the greatest advantages of multitask learning is that only one model is needed to simultaneously accomplish the prediction for the 12 targets. In addition, the scale of the model parameters as reduced by decreasing the shared representation, so training efficiency was greatly improved. With the help of the weight sharing from multitask learning, the convergence rate of multitask CapsNet was greatly improved. Overall, multitask

learning alleviated the dilemma of a serious shortage of positive molecular data for certain targets through transfer learning between tasks, thereby improving the generalization ability of the model.

CONCLUSIONS

In this investigation, a novel deep learning method, called multitask CapsNet, was developed for simultaneously predicting 12 toxic effects. In addition, four conventional deep learning methods were used to develop multitask prediction models for toxicities. Among these models, our multitask CapsNet presented the best predictive performance, producing the largest ratio (8/12) of the highest overall accuracies for the

leaderboard data set as well as the highest prediction accuracies of toxicants for all 12 targets. Compared with other previous approaches under the same conditions, our model performed equally well or even superior. Notably, our method exhibited good predictive ability for biased data sets without manually engineered data and features, and it optimized the loss faster and converged to a final lower loss faster than multitask DBN. The results collectively demonstrated that the adverse effects resulting from bias in data sets can be relieved with the help of multitask CapsNet; thus, this novel method provided a promising alternative for the qualitative prediction of various compound properties in the early stages of drug research.

MATERIALS AND METHODS

Data Preparation. The Tox21 program is a federal collaboration between the National Institute of Environmental Health Sciences/National Toxicology Program, the U.S. Environmental Protection Agency/National Center for Computational Toxicology, the National Institutes of Health Chemical Genomics Center, and the U.S. Food and Drug Administration. Tox21 aims to develop better toxicity assessment methods to quickly and efficiently test whether certain chemical compounds have the potential to disrupt processes in the human body that may lead to negative health effects. Because it originated from standard experimental conditions with low noise levels, Tox21 has become a benchmark data set and is widely used in toxicity prediction modeling. The Tox21 data set contains more than 10 000 compounds on 12 different toxicities. The endpoints are given a binary label 1 or 0, with 1 representing toxicants and 0 nontoxicants. The original Tox21 data set is divided into a training data set, a leaderboard set, and a test set, each set comprising 11 764, 296, and 647 compounds, respectively.^{30,50} The training and leaderboard sets are generally used to train and evaluate the performance of different computational methods for the 12 targets. As shown in Table 2, the Tox21 data set is rather biased, as it includes a larger number of negatives (nontoxicants) than positives (toxicants), with an imbalance ratio of positive to negative compounds ranging from approximately 1:5 to 1:36.

Calculation of Molecular Features. Molecular fingerprints, together with properties, were used simultaneously as descriptors for the input of the models. MDL Molecular Access (MACCS) molecular fingerprint is a class of one-dimensional bit arrays, which can be directly recognized and learned by computers. Compared with other molecular fingerprints, the MACCS fingerprint (166 bits) has a moderate length and contains important information on the molecular structure.⁵¹ Using the MACCS fingerprint, the multitask CapsNet model would have speedy dimension reduction while maintaining complete information on the molecule structure. Thirteen types of molecular properties were used in our study, and the detailed list of these properties is given in Table 3. These molecular properties have been widely used in many aspects of computer-aided drug design, such as virtual screening, similarity search, and constructing models for ADMET in pharmacokinetic predictions. The MACCS fingerprint was computed using PaDEL-Descriptor,⁵² and molecular properties were calculated using RDKit (version 2018.09.1).⁵³ The values of various descriptors span different numerical areas, so we scaled their values to the same range (0,1) using the following formula

Table 3. Summary of the 13 Types of Molecular Properties Used

no.	molecular properties
1	octanol–water partitioning coefficient
2	apparent partition coefficient at pH = 7.4
3	molecular solubility
4	molecular weight
5	number of hydrogen bond donors
6	number of hydrogen bond acceptors
7	number of rotatable bonds
8	number of rings
9	number of aromatic rings
10	sum of the oxygen and nitrogen atoms
11	polar surface area
12	molecular fractional polar surface area
13	molecular surface area

$$x^* = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

where x is the initial value of the descriptor, x^* is the scaled value of the descriptor, and $x_{\max}$ and $x_{\min}$ correspond to the maximum and minimum values of each class of descriptor, respectively.

CapsNet architectures are end-to-end models, which can learn features automatically by the dynamic routing algorithm; they do not require prior feature selection. Moreover, we tried to explore how different descriptors as model inputs influence the performance of models in previous research, and the results indicated that different descriptors did not have a significant impact on the models.⁵⁰ Therefore, it is reasonable to adopt all of the 179 molecular features to build the novel models.

Network Architecture of the Multitask CapsNet Model. To achieve the best model, five network architectures integrating with one to five RBM operations were explored. The candidate architecture with two RBMs that provided the highest accuracy is shown in Figure 6. This model consists of four layers: an input layer, a shared hidden feature layer, a PrimaryCaps layer, and a DigitCaps layer (output layer). The input layer receives the input from the 12 tasks simultaneously. The original input is the one-dimensional vector, which comprises 166 bits of MACCS fingerprints and 13 molecular descriptors, so the input layer possesses 179 neurons. Two RBMs make up the shared hidden feature layer, in which the first and second RBMs contain 256 and 128 neurons, respectively. The first RBM (RBM1) receives and encodes the original input from the input layer, and the vectors are transmitted to the second RBM (RBM2).

The origin architecture of CapsNet was proposed by Hinton et al., which contains three layers: a convolutional layer, a PrimaryCaps layer, and a DigitCaps layer, and each layer consists of a group of neuron vectors called a capsule.^{47,54} In our model, the PrimaryCaps layer comprises eight capsules and each capsule includes eight-dimensional features. In the DigitCaps layer, the two classification capsules are generated with two dynamic routing iterations between the PrimaryCaps and DigitCaps layers. The final layer (DigitCaps) has two capsules, each of which is a two-dimensional capsule. The outputs of the DigitCaps layer predict the labels corresponding to the nontoxics and the toxics. We predicted the class of chemical molecules by computing the length of each digit capsule. In view of the small data set in this study, we added

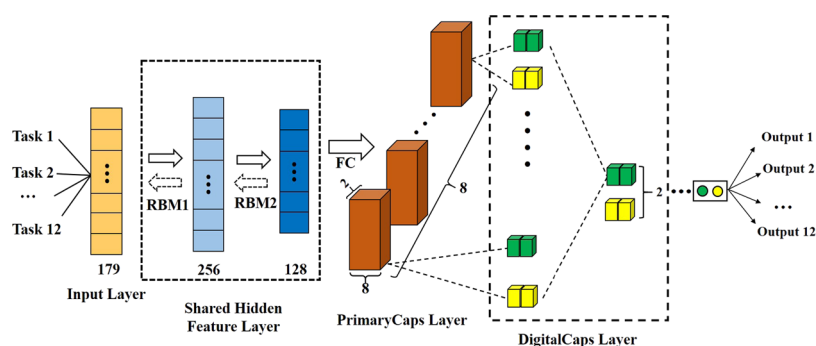


Figure 6. Architecture of multitask CapsNet. Multitask CapsNet with four layers. The input layer is a one-dimensional vector consisting of 179 components. The shared hidden feature layer is made up of two RBMs; RBM1 and RBM2 include 256 and 128 nodes, respectively. The PrimaryCaps layer contains 8×8 neural units. The DigitCaps layer contains eight neural units. Each task shares the learning process of molecular representation and has its own four layers for building the corresponding predictor of its toxicity property.

the dropout technique before the activation function to prevent the network from overfitting.⁵⁵ Detailed information related to the equations for each layer in multitask CapsNet is listed in the [Supporting Information](#).

As training each subnetwork from scratch is very time-consuming, many works leverage weight sharing to speed up the training process.⁵⁶ Inspired by this, we employed the idea of weight sharing over multiple scales in multitask CapsNet. Training RBMs is extremely laborious since this algorithm often converges only with the right set of learning rates and the scale of the initial weights. Hence, we utilized parameter sharing in the shared hidden feature layer, namely, all of the subnetwork shared the same parameters in this layer.

Model Training of Multitask CapsNet. The training of multitask CapsNet contains two stages: pretraining and fine-tuning. The pretraining process can be implemented layer-by-layer from low to high layers to train the two RBMs. In the fine-tuning step, the loss function presented by Hinton was employed to achieve end-to-end optimization in the RBM layer, and the contribution matrix was generated via a back-propagation algorithm. The detailed algorithm and training procedure of the multitask CapsNet model is summarized in [Table S8](#).

Construction of the Compared Deep Learning Models. Four commonly used deep learning architectures, deep belief network (DBN), convolutional neural network (CNN), long short-term memory (LSTM), and multilayer perceptron (MLP), were integrated with the multitask learning strategy and applied for comparison with multitask CapsNet for each data set of Tox21. For multitask DBN, eight hyperparameter settings were optimized, and the model with the two RBMs was selected for the same RBM composition as that of multitask CapsNet. For multitask CNN, we optimized seven hyperparameter settings, and the model was stacked with three one-dimensional convolutional layers and one fully connected layer. For multitask LSTM, we employed seven different hyperparameters to optimize the model. For multitask MLP, four hyperparameters were used to obtain the best model. The default values were applied, as the hyperparameters were not detailed. All of the deep learning models were developed by TensorFlow 1.10.0. The hyperparameter space considerations for these four models and optimized hyperparameter values are listed in [Tables S9–S12](#).

Performance Evaluation of the Models. Each model was assessed for the qualities of accuracy (Q), sensitivity (SE), and specificity (SP). Accuracy merely reflects the total

predicted effect of the classifier, while sensitivity and specificity represent the predictive power of the model for positives and negatives, respectively. These evaluation indexes were defined as follows

$$Q = \frac{TP + TN}{TP + FP + TN + FN} \quad (1)$$

$$SE = \frac{TP}{TP + FN} \quad (2)$$

and

$$SP = \frac{TN}{FP + TN} \quad (3)$$

Another measure of the quality of binary (two-class) classifications is Matthew's correlation coefficient (MCC). MCC considers the balance ratios of the four confusion matrix categories (true positives, true negatives, false positives, and false negatives), so it can reflect the prediction power of models objectively without the influence of a disproportionate ratio of positives to negatives in the data set. MCC was calculated by the following equation

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(FP + TN)(FP + TP)(FN + TN)(FN + TP)}} \quad (4)$$

The area under the curve (AUC) is an important indicator of classifiers, which is defined by the receiver operating characteristic curve (ROC). Different from other models, the predictive output of the capsule is determined by a comparison of the lengths of the classified capsules.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsomega.1c03842>.

Detailed information related to the equations for each layer in multitask CapsNet; selection of the optimal multitask CapsNet; detailed performances of different multitask models developed by five deep learning algorithms; comparison of multitask CapsNet with other models in the DeepTox pipeline; detailed performances of multitask CapsNet and multitask DBN models training on balanced sets; algorithm and

training procedure of the multitask CapsNet model; and optimal hyperparameter settings of the other four multitask deep learning models (PDF)

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Notes

The authors declare no competing financial interest.

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